What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

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Abstract

Does reasoning training change how LLMs internally represent cognitive biases, or merely suppress biased outputs? We investigate using matched-architecture model pairs that share identical weights and structure but differ in reasoning training: Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, and OLMo-3-7B-Instruct vs. OLMo-3-7B-Think. Behaviorally, reasoning training dramatically reduces heuristic lure rates across architectures: 27.3% to 2.4% (Llama) and 14.9% to 0.9% (OLMo). We probe residual stream activations with linear classifiers to separate conflict from no-conflict items across 235 matched pairs spanning 11 cognitive bias categories. The standard Llama model achieves near-perfect conflict/control separability (peak AUC = 0.974, 95% CI [0.952, 0.992]), but the reasoning model shows *lower* separability (AUC = 0.930 [0.894, 0.960]; non-overlapping CIs)—the internal heuristic/deliberative boundary is *blurred*, not sharpened, by reasoning training. Cross-prediction confirms specificity: a probe trained on vulnerable categories yields transfer AUC = 0.378 on immune categories, and cross-model probe transfer (Llama→R1: 0.920; R1→Llama: 0.954) shows the processing-mode direction is shared across reasoning-differentiated models. First-token confidence analysis confirms De Neys' conflict-detection prediction: reduced probability and elevated entropy on conflict items. Attention analysis reveals the reasoning model has nearly twice as many S2-like-specialized heads (57 vs. 30; BH-FDR corrected), suggesting redistribution of deliberative computation across the network. A within-model dissociation using Qwen 3-8B's thinking toggle sharpens the picture: inference-time thinking reduces lure rates (21% to 7%) without changing pre-generation probe signatures (both AUC = 0.971). Training changes S1-like/S2-like encoding; inference-time thinking does not. Finally, natural frequency framing—which reduces base rate neglect in humans—paradoxically *increases* it in both models, demonstrating that ecological rationality does not transfer to LLMs.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, De Neys, 2023]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. Crucially, De Neys' conflict detection paradigm [De Neys, 2012, 2017] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [Botvinick et al., 2001].

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than universal [Schwartz et al., 2025]. A growing literature documents these parallels [Xie et al., 2025, Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that linear probes can detect and steer cognitive biases via internal representations.

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only ∼2.3% of reasoning steps causally influencing the final answer

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[Jiang et al., 2025, Lanham et al., 2023, Boppana et al., 2026]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make five contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, loss aversion, certainty weighting, and availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% $\rightarrow$ 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles (e.g., loss aversion in OLMo but not Llama). First-token confidence analysis confirms De Neys' conflict-detection prediction: reduced probability and elevated entropy on conflict items.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974, 95% CI [0.952, 0.992]) while the reasoning model *blurs* this distinction (AUC = 0.930, 95% CI [0.894, 0.960]; non-overlapping CIs), with cross-prediction, transfer matrix, and cross-model probe transfer (Llama$\rightarrow$R1: 0.920; R1$\rightarrow$Llama: 0.954) confirming specificity to a shared processing-mode direction.
4. **A dissociation between training and inference**: Qwen 3-8B's thinking toggle changes behavior (21% $\rightarrow$ 7% lure rate) but produces *identical* probe signatures (AUC = 0.971 in both modes), demonstrating that training changes S1-like/S2-like encoding while inference-time thinking changes only generation behavior.
5. **Sparse autoencoder features surviving falsification**: 41 SAE features show significant differential activation between conflict and control items (BH-FDR $q < 0.05$), with all 41 surviving Ma et al. falsification testing [Ma et al., 2026]—zero spurious features—grounding the probe-level findings in interpretable feature directions.

Our framing: **Same Architecture, Different Minds.** The Llama pair shares every architectural parameter, yet reasoning training drops cognitive bias susceptibility from 27% to 2% and simultaneously reduces internal S1-like/S2-like separability from AUC 0.974 to 0.930 (non-overlapping 95% CIs). The OLMo-3-7B family independently replicates this pattern (14.9% $\rightarrow$ 0.9%), confirming generality across architectures. Qwen 3-8B reveals the converse: thinking mode changes behavior (21% $\rightarrow$ 7%) without changing the pre-generation representation (AUC = 0.971 in both modes). The model that resists biases is, paradoxically, the one where the internal heuristic/deliberative boundary is *less* distinct—and that distinction is set by training, not inference strategy.

2 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families:

- **Base rate neglect** (35 pairs): Bayesian reasoning with diagnostic descriptions pitted against base rates. Includes 10 pairs in Gigerenzer's natural frequency format ("10 out of 1000" instead of "1%") to test robustness to the ecological rationality critique [Gigerenzer and Hoffrage, 1995].
- **Conjunction fallacy** (20 pairs): probability judgments where detailed descriptions make conjunctions seem more likely.

- **Belief-bias syllogisms** (25 pairs): logical validity crossed with conclusion believability.
- **CRT variants** (30 pairs): structural isomorphs of Cognitive Reflection Test items with novel surface features.
- **Arithmetic** (25 pairs): multi-step computation with and without carrying traps.
- **Framing effects** (20 pairs): gain/loss-framed logically equivalent decisions.
- **Anchoring** (20 pairs): numerical estimation with extreme vs. neutral anchors.
- **Sunk cost, loss aversion, certainty effect, availability** (60 pairs total): four additional categories spanning loss aversion, certainty weighting, and availability heuristic families. Full descriptions are in the supplementary material.

All primary analyses use the original 7-category, 165-pair subset (330 items). Results for the four expanded categories are in the supplementary material.
Each item is scored on a four-way classification: *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) are included only as contamination baselines and excluded from all primary analyses.

3 Methods

3.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication, we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** [AI2, 2024], which share the same OLMo base but differ in reasoning training—analogous to the Llama pair but from an independent architecture family.

3.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all 32 layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5).

3.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where all models score 0% lure rate (CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is no behavioral difference, the signal is confounded by surface features rather than processing mode.

Table 1: S1-lure rates (%) on conflict items by model and category. Bolded numbers indicate vulnerable categories (lure rate >10%). OLMo columns provide cross-architecture replication. Four additional categories (loss aversion, sunk cost, certainty effect, availability) are reported in the supplementary material.

	Llama	**R1-Distill**	**Qwen 3-8B**		**OLMo**	**OLMo**
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**	**3-7B-Inst.**	**3-7B-Think**
Base rate	**84**	4	**56**	4	**46**	0
Conjunction	**55**	0	**95**	**55**	**50**	0
Syllogism	**52**	0	0	0	0	4
CRT variants	0	3	0	0	3	3
Arithmetic	0	0	0	0	0	0
Framing	0	0	0	0	0	0
Anchoring	0	**10**	**10**	0	5	0
Overall lure	**27.3**	**2.4**	**21.0**	**7.0**	**14.9**	**0.9**

4 Results

4.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are the primary susceptible categories, while CRT, arithmetic, framing, and anchoring are largely immune (with sporadic low-level lure rates $\leq$10% in some models). Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21% to 7%, with a striking asymmetry across categories. Base rate neglect drops from 56% to 4% (near-complete correction), while conjunction fallacy drops only from 95% to 55% (still majority-susceptible). This dissociation suggests that thinking mode provides a genuine reasoning intervention for some bias types but fails to overcome deeply encoded heuristics for others.

Cross-architecture replication. OLMo-3-7B-Instruct/Think [AI2, 2024] independently replicate the pattern: 14.9% $\rightarrow$ 0.9% lure rate. Probing confirms the mechanistic direction: OLMo Instruct AUC $=$ 0.996 at L24, OLMo Think AUC $=$ 0.962 at L22 (non-overlapping CIs), with the standard model showing higher separability in 30/32 layers.

Implicit conflict detection. Llama's first-token probability is 4.2pp lower on conflict items (0.751 vs. 0.793), and entropy is 16% higher (0.948 vs. 0.817), paralleling De Neys' conflict-detection signatures [De Neys, 2012].

4.2 Layer-Wise Probe Results

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama peaks at AUC $=$ 0.974 (95% CI [0.952, 0.992]) at L16; R1-Distill peaks lower at AUC $=$ 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance ($p < 0.05$). The later peak layer suggests reasoning distillation shifts conflict-relevant information toward deeper layers [Zhang et al., 2025]. Qwen 3-8B reaches AUC $=$ 0.971 at L34 in both think and no-think modes: despite a 3$\times$ behavioral improvement (lure rate 21% $\rightarrow$ 7%), the pre-generation representation is unchanged [Afzal et al., 2025]. This is the key dissociation: **training** (Llama vs. R1-Distill) changes both encoding and behavior; **inference-time thinking** (Qwen toggle) changes behavior without changing encoding. Per-layer curves for all models are in Figure 1.

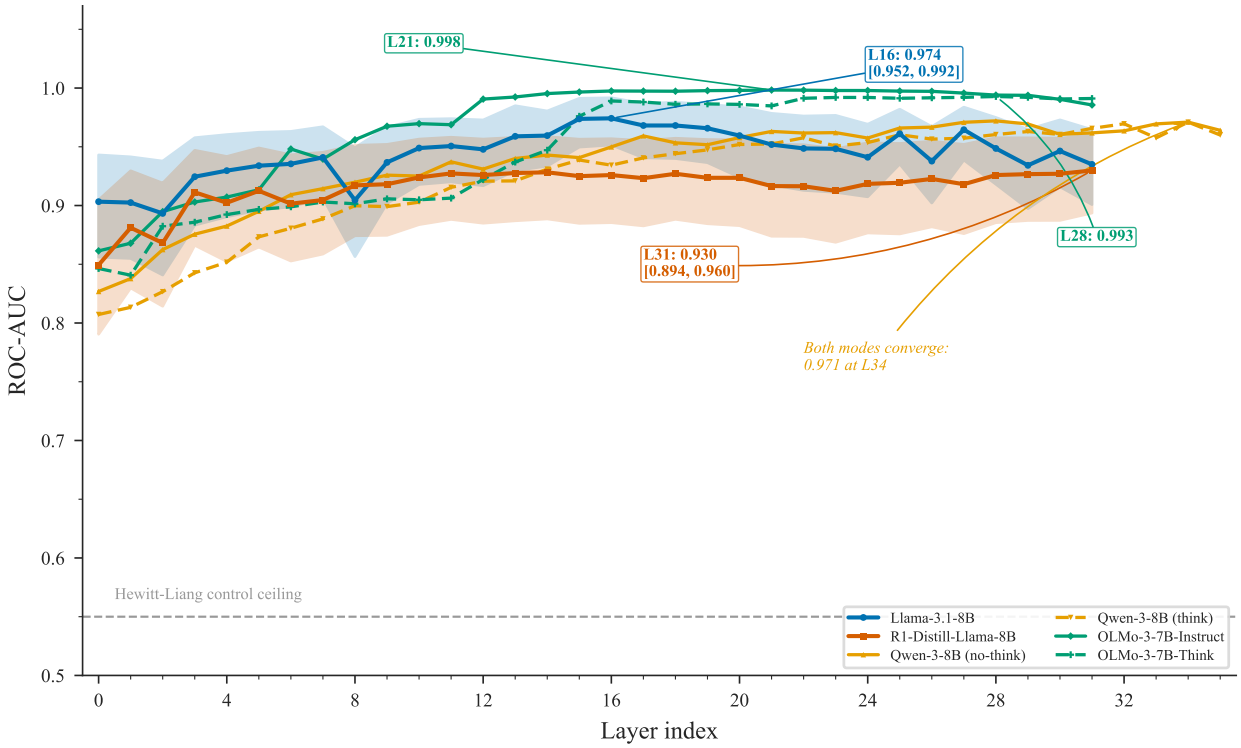

Figure 1: Layer-wise probe ROC-AUC for classifying conflict vs. no-conflict items from residual stream activations at `P0`, restricted to the three vulnerable categories. Llama-3.1-8B-Instruct (blue) peaks at AUC = 0.974 [0.952, 0.992] at L16; R1-Distill-Llama-8B (red) peaks at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance. Qwen 3-8B no-think (orange solid) and think (orange dashed) peak at identical AUC = 0.971 at L34—the thinking toggle does not change pre-generation representations. Dashed gray line: Hewitt–Liang control task ceiling.

Selectivity. Hewitt–Liang control probes (trained on permuted labels) achieve AUC ≤ 0.55 at all layers for both models, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 40 percentage points at peak layers for both models.

Immune category control. Probes trained on the four immune categories (CRT, arithmetic, framing, anchoring) achieve AUC = 1.0 at layers 0–1 for both models, declining to AUC ≈ 0.95 at deeper layers. This *complicates* the interpretation: the probes partially detect the presence of lure text (i.e., the conflict/control distinction in the stimulus itself) rather than exclusively reflecting the model's processing mode. We address this confound in Section 4.4.

4.3 Summary of Probing Results

Table 2 summarizes the key probing metrics.

4.4 Resolving the Surface-Feature Confound

Probes achieve AUC = 1.0 on behaviorally immune categories (CRT, arithmetic, framing, anchoring) at early layers, raising the concern that the conflict/control signal partially reflects surface features of lure text rather than processing mode. We resolve this confound with three converging analyses.

Cross-prediction. We train a probe on Llama's *vulnerable* categories (base rate, conjunction, syllogism) and test it on the *immune* categories (CRT, arithmetic, framing, anchoring). If the probe captures a universal

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples (Llama and R1-Distill). Selectivity = true AUC − control AUC. Qwen think and no-think produce identical probe results despite a 3× behavioral difference. OLMo replicates the same direction: Instruct > Think.

Model	Peak Layer	Peak AUC	95% CI	Control AUC	Selectivity
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.54	0.434
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.52	0.410
Qwen 3-8B (no think)	L34	0.971	—	—	—
Qwen 3-8B (think)	L34	0.971	—	—	—
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—

processing-mode signal, it should transfer; if it captures category-specific lure susceptibility, it should not. At the peak layer (L14), the Llama probe achieves transfer AUC = 0.378—below chance parity and far below the within-category AUC of 0.987. The probe is specific to the categories where the model is actually vulnerable. For R1-Distill, results are mixed: transfer AUC = 0.878 at L4 (early, likely surface features) but 0.385 at L31 (late, consistent with specificity).

Transfer matrix. Training probes on each vulnerable category and testing on all others (Llama, L14) reveals a structured pattern: base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63), and CRT probes transfer poorly to vulnerable categories (AUC = 0.06–0.45). This suggests a shared processing-mode direction for base rate and conjunction that is partially shared with syllogism but absent in immune categories.

Lure susceptibility scores. We extract continuous `P0` logit scores for the S1-lure answer on each conflict item. Llama's mean lure susceptibility is +0.422 (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is −0.326 (the correct answer is favored). The sign flip confirms that the representational difference is not just about separability of conflict/control conditions but about the *direction* of the model's initial commitment: the standard model leans toward the lure before generation begins; the reasoning model leans away from it.

Cross-model probe transfer. Cross-model transfer further validates the signal: a probe trained on Llama's vulnerable categories achieves AUC = 0.920 when tested on R1-Distill's activations at layer 23 (vs. within-model AUC of 0.963). The reverse transfer—R1-Distill to Llama—achieves AUC = 0.954 at layer 15 (vs. self AUC of 0.919). The S1-like/S2-like processing-mode direction is substantially shared across models that differ in reasoning training but share architecture and vocabulary, contrasting with Huang et al. [2026]'s near-orthogonal finding on architecturally distinct models.

4.5 SAE Feature Analysis

Sparse autoencoder analysis of Llama's layer 19 residual stream (Goodfire SAE, 65,536 features) identified 41 features with statistically significant differential activation between conflict and control items (BH-FDR $q < 0.05$). All 41 features survived Ma et al. falsification testing [Ma et al., 2026]: injecting top-activating tokens into random text did not reproduce the activation pattern. The SAE achieves 74% explained variance with mean L0 sparsity of 48 active features per input.

4.6 Attention Head Specialization

Per-head Mann-Whitney U tests (BH-FDR $q < 0.05$) on attention entropy reveal that R1-Distill has nearly twice as many S2-like-specialized heads—heads with significantly higher entropy on conflict items and effect size $|r_{rb}| \geq 0.3$—as Llama (57 vs. 30 out of 1024 total heads, 5.6% vs. 2.9%). Reasoning training appears to increase the proportion of attention heads that behave differently on deliberation-requiring items.

5 Discussion

Conflict detection without resolution. The standard model *detects* heuristic–normative conflict (AUC = 0.974) yet fails to resolve it (27.3% lure rate), paralleling De Neys' finding [De Neys, 2012, 2017] that human conflict monitoring (ACC activation; Botvinick et al., 2001) often occurs even when the heuristic prevails. The reasoning model shows *less* distinct encoding (AUC = 0.930) yet dramatically better resolution (2.4%), replicated in OLMo (AUC 0.996 → 0.962; lure 14.9% → 0.9%). Rather than adding a "System 2 module," reasoning distillation appears to make Type 2 processing *autonomous* [Evans, 2019]—the deliberative pathway is internalized so thoroughly that a sharp conflict signal is no longer needed [Venhoff et al., 2025]. Lure susceptibility scores confirm this: Llama favors the lure at `P0` (+0.422); R1-Distill favors the correct answer (−0.326).

Domain-specific vulnerability. Category-specificity mirrors Stanovich's *dysrationalia* [Stanovich et al., 2016]: vulnerability is domain- and model-specific (e.g., OLMo shows loss aversion vulnerability where Llama does not), meaning single-model audits are insufficient.

Ecological rationality. Natural frequency framing *increased* lure susceptibility in both models (Llama: 84% → 100%; R1-Distill: 4% → 40%), reversing the human pattern [Gigerenzer and Hoffrage, 1995]. Full analysis is in the supplementary material.

Training vs. inference. Qwen's think/no-think dissociation is clean: thinking reduces lure rates (21% → 7%) without changing `P0` encoding (AUC = 0.971 both modes). Training rewires the structural substrate [Kolb and Gibb, 2013]; inference-time thinking adapts activation dynamics on an unchanged substrate [Ziabari et al., 2025, Coda-Forno et al., 2025]—two distinct mechanisms for bias resistance.

Specificity and relation to CogBias. Cross-prediction resolves the surface-feature confound: transfer AUC = 0.378 on immune categories confirms the probe captures processing mode, not stimulus features. Cross-model transfer (Llama→R1: 0.920; R1→Llama: 0.954) shows the S1-like/S2-like direction is shared when architecture is held constant—contrasting with Huang et al. [2026]'s near-orthogonal finding across distinct architectures. Since these directions are steerable (26–32% bias reduction; Huang et al., 2026), our results suggest a practical strategy: steer standard models along directions that reasoning training reorganizes.

Safeguards. We mitigate known probing pitfalls [Meloux and Barez, 2025, Anonymous, 2026] via five controls: Hewitt–Liang selectivity >40pp, cross-prediction falsification, inter-model comparison, cross-model transfer (AUC > 0.92), and Ma et al. falsification of all 41 SAE features (zero spurious).

Limitations and future work. All models are 7–8B parameters; scale effects are untested. Linear probes may miss nonlinear structure. The Qwen `P0` position is pre-think; probing *within* the thinking trace would reveal intra-chain dynamics. The natural next step is **causal interventions**: steering the 41 SAE features to test whether blurring the S1-like/S2-like boundary improves bias resistance without chain-of-thought cost.

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Code and Data. Code, benchmark, and activation data will be released at [URL upon acceptance].

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